

WIRELESS AND MOBILE AD HOC NETWORKS

Complete Exam-Oriented Notes

Based strictly on the provided syllabus

Unit I	Basics of Networking
Unit II	Mobile Ad Hoc Network (MANET)

Designed for university examination preparation: concepts → working → examples → exam points → quick revision.

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Final Final Revision Section

Syllabus coverage

Every topic supplied in the prompt is explicitly represented in the notes. The final section consolidates definitions, formulas, comparisons, diagrams/concepts, testable points, and rapid unit revision.

Unit I — Basics of Networking

Unit focus

This unit builds the foundation needed to understand wireless and mobile networking. It covers networking concepts, wireless components and evolution, topologies, switching, protocols, addressing, internetworking, interdomain routing, and practical network-analysis tools.

1.1. Introduction

A **computer network** is a collection of interconnected devices that communicate and share data, applications, and resources using agreed communication rules called protocols. A network may connect a few devices in a room or millions of devices across the Internet.

Wireless networking transfers information using electromagnetic waves instead of a physical cable for the wireless portion of the connection. Wi-Fi, Bluetooth, cellular networks, satellite links, and many sensor/IoT systems are examples.

Why networking matters: networking enables resource sharing, communication, distributed applications, remote access, scalability, and centralized or distributed services.

Basic communication sequence

- 1 A source creates data.
- 2 Data is encapsulated into protocol data units as it moves down the protocol stack.
- 3 The physical/wireless medium carries signals.
- 4 Intermediate devices forward traffic according to addressing and routing information.
- 5 The destination decapsulates the data and delivers it to the application.

Key networking terms

Term	Meaning
Node	Any networked device that can send, receive, or forward data.
Link	A communication path between two nodes.
Packet	A network-layer unit of data, typically containing a header and payload.
Protocol	A set of rules defining communication format, meaning, and behavior.
Bandwidth	The capacity/rate of a communication link, commonly expressed in bit/s.
Latency	Time taken for data to travel from source toward destination.
Throughput	Actual useful data delivered per unit time.

Quick Revision

Network = communicating nodes + links + protocols. Wireless networking replaces at least part of the wired medium with radio, microwave, infrared, or similar wireless transmission.

1.2. Types of Wireless Networking Components

Wireless networks are built from **end devices, wireless access infrastructure, forwarding devices, transmission media, and control/management components**. The exact components depend on the network type.

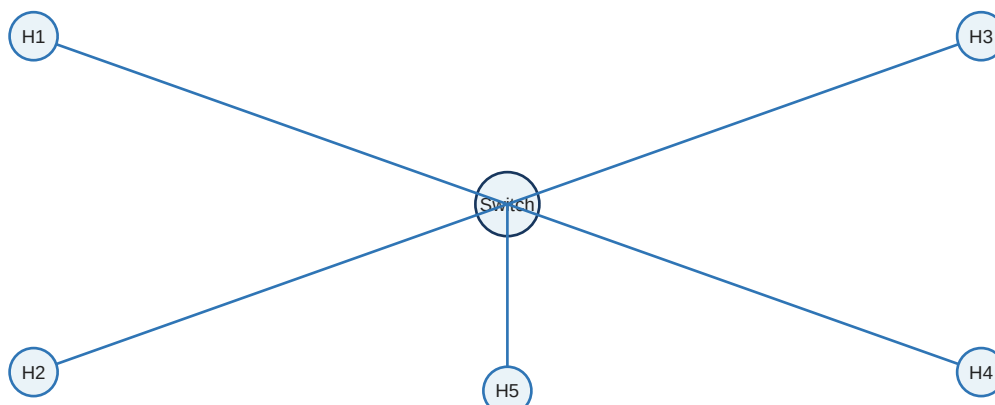
Component	Role	Typical example
Wireless client / station	Originates or receives wireless traffic.	Laptop, smartphone, sensor
Access Point (AP)	Connects wireless stations to a distribution network; coordinates WLAN access.	WLAN AP
Wireless router	Combines routing, switching, and often wireless access functions.	Home/office router
Base station	Provides radio coverage and connects mobile users to a cellular network.	Cellular base station
Router	Forwards packets between IP networks using routing information.	Enterprise/Internet router
Switch	Forwards frames inside a LAN, usually using MAC addresses.	Ethernet switch
Gateway	Connects systems/networks using different protocols or administrative domains.	Network gateway
Wireless medium	Carries signals through radio or other wireless propagation.	2.4/5 GHz radio
Controller/management system	Configures, monitors, authenticates, or coordinates infrastructure WLAN.	WLAN controller / management server

A useful distinction is **infrastructure-based** networking versus **ad hoc** networking. In infrastructure mode, stations normally communicate through fixed access points or base stations. In an ad hoc network, nodes can communicate directly and may also forward traffic for one another.

Exam Point

Do not confuse an AP with a router. An AP primarily provides wireless LAN connectivity; a wireless router commonly combines AP, switching, routing, DHCP/NAT, and other functions.

Diagram: simple infrastructure WLAN



Quick Revision

Remember the main roles: station = endpoint; AP/base station = wireless access; switch = LAN frame forwarding; router = inter-network packet forwarding; gateway = protocol/network boundary.

1.3. Evolution of Wireless Technologies

Wireless technology evolved from basic short-range radio systems to high-speed, low-latency, highly mobile broadband and IoT systems. Each generation generally improved capacity, coverage, mobility, reliability, spectrum efficiency, and supported applications.

Era / technology family	Main idea	Typical contribution
Early radio / 1G cellular	Analog voice and basic mobile communication	Mobile voice service.
2G	Digital cellular communication.	Better capacity, SMS, digital voice; packet data emerged.
3G	Higher-rate mobile packet data.	Mobile Internet and multimedia services.

Era / technology family	Main idea	Typical contribution
4G LTE	All-IP mobile broadband.	High data rates, lower latency, IP multimedia.
5G	High-capacity, flexible radio/network architecture.	Enhanced broadband, massive IoT, low-latency use cases.
Wi-Fi evolution	IEEE 802.11 family progressively increased throughput, efficiency and features.	High speed, efficiency and features.
Bluetooth evolution	Short-range personal-area networking with improved power and data capabilities.	Peripheral device connectivity.
LPWAN / IoT technologies	Long-range, low-power communication for small-scale sensors and IoT devices.	Long range, low power.

Exam idea: evolution is not simply “newer = faster.” Newer systems also target efficiency, density, power consumption, reliability, latency, spectrum use, and support for diverse devices.

Quick Revision

1G → analog mobile voice; 2G → digital voice/SMS; 3G → mobile data; 4G → broadband all-IP; 5G → high capacity + flexible services + massive/low-latency scenarios. Wi-Fi and Bluetooth evolved in parallel for local/personal connectivity.

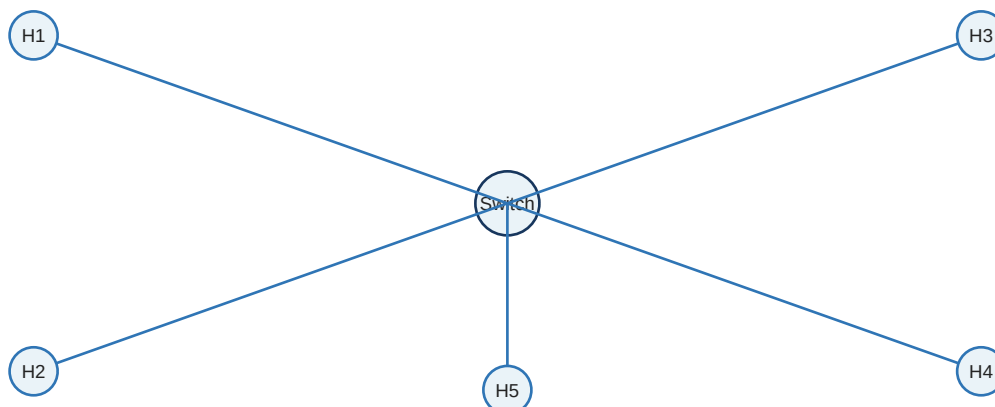
1.4. Network Topologies

A **network topology** describes how nodes and links are arranged. It may refer to physical placement or logical communication relationships.

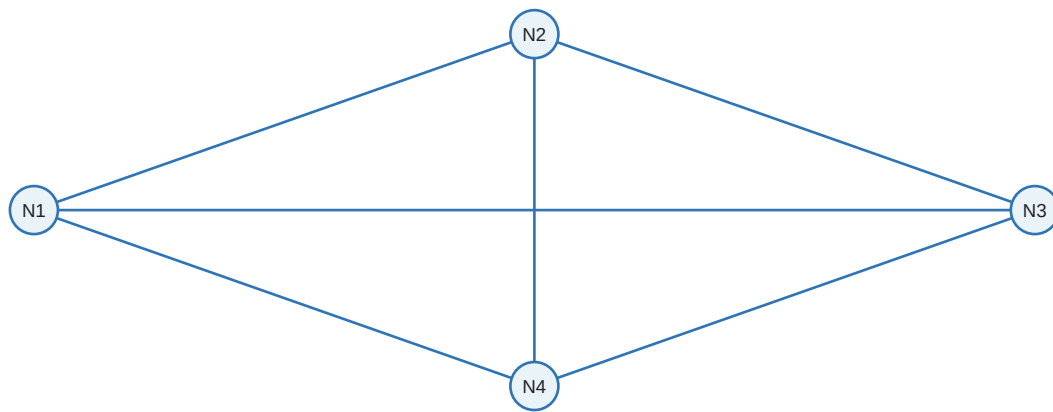
Topology	Structure	Advantages	Limitations / exam point
Bus	All nodes share a common backbone.	Simple, low cabling in small networks.	Backbone failure can affect many nodes; shared medium.
Star	All nodes connect to a central device.	Easy management; a link failure usually affects one node only.	Central device is a critical point.
Ring	Each node connects to two neighbours in a single loop.	Predictable structure; can support orderly access.	A break can disrupt a simple ring unless redundancy exists.
Mesh	Nodes have many/all possible interconnections.	High redundancy and fault tolerance.	Expensive and complex as node count grows.
Tree / hierarchical	Stars are organized in levels.	Scalable and easy to segment.	Higher-level failure can affect branches.
Hybrid	Combination of two or more topologies.	Flexible and practical for large networks.	More complex design.

Wireless networks often use logical or radio connectivity rather than a literal cable layout. A MANET may have a **dynamic graph topology**: links appear and disappear as nodes move or radio conditions change.

Star topology



Mesh topology



Quick Revision

Star = central point. Mesh = many redundant paths. Ring = closed loop. Bus = shared backbone. In MANETs, topology is often dynamic because mobility changes which nodes can reach each other.

1.5. Switching Techniques

Switching is the method used to move data through a network from source toward destination. The major classical techniques are circuit switching, packet switching, and message switching.

Technique	How it works	Strength	Weakness / use
Circuit switching	A dedicated end-to-end path/resources are established before data transfer.	Predictable resource allocation.	Setup overhead; inefficient for bursty data.
Packet switching	Data is split into packets; network resources are shared dynamically.	Efficiently handles computer traffic.	Variable delay, queuing, packet loss possible.
Message switching	Entire message is stored and forwarded at intermediate nodes.	No dedicated circuit required.	Large storage and delay; rarely used for modern networks.

Packet switching models: in a **datagram network**, each packet is routed independently; in a **virtual-circuit approach**, a logical path/state is established and packets follow that association.

Why packet switching fits the Internet

- Internet traffic is bursty rather than continuous.
- Many users can share link capacity statistically.
- Packets can be routed around failures or congestion.
- Different applications can coexist on the same infrastructure.

Quick Revision

Circuit switching reserves a path; packet switching shares network resources packet-by-packet; message switching stores complete messages. Modern IP networks are packet-switched.

1.6. Network Protocols

A **network protocol** is a formal set of rules that governs communication. Protocols define how information is formatted, addressed, transmitted, acknowledged, routed, and interpreted.

Important protocol families

Protocol / family	Layer / role	Key purpose
Ethernet (IEEE 802.3)	Link	LAN framing and media access.
Wi-Fi (IEEE 802.11)	Link/physical	Wireless LAN communication.

Protocol / family	Layer / role	Key purpose
IP (IPv4/IPv6)	Network	Logical addressing and packet forwarding.
ARP / Neighbor Discovery	Link/network support	Resolve local next-hop link-layer information.
TCP	Transport	Reliable, ordered byte-stream delivery with congestion control.
UDP	Transport	Connectionless datagram delivery with low overhead.
DNS	Application	Maps names to network addresses and related records.
DHCP	Application	Automates host network configuration.
HTTP/HTTPS	Application	Web resource communication.
BGP	Interdomain routing	Exchanges reachability information between autonomous systems.

Encapsulation: application data is wrapped by transport information, then network-layer information, then link-layer information. At the receiver, the process is reversed (decapsulation).

Exam Point

A protocol is not the same as a device. TCP, IP, HTTP and BGP are protocols; router, switch and AP are devices/functions that use protocols.

Quick Revision

Protocols define communication rules. TCP focuses on reliable transport; UDP on lightweight datagrams; IP on addressing/forwarding; DNS on naming; DHCP on configuration; BGP on interdomain reachability.

1.7. IP Addressing

IP addressing provides logical identifiers that allow devices/interfaces to be located and packets to be forwarded across interconnected networks. IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses.

IPv4 notation and subnetting

IPv4 address = 32 bits, normally written as four decimal octets. A subnet mask/prefix length divides the address into a **network prefix** and **host/interface portion**.

Example: 192.168.1.0/24

Here, /24 means the first 24 bits are the network prefix and 8 bits remain for host addresses. For a conventional /24 subnet, 192.168.1.0 is the network address and 192.168.1.255 is the broadcast address; usable host addresses are typically .1 through .254.

Useful IPv4 formulas

Number of addresses in an IPv4 subnet = $2^{(32 - \text{prefix length})}$. For traditional subnetting, usable host addresses are often $2^h - 2$, where h is the number of host bits.

Address type	Purpose
Network address	Identifies the subnet itself.
Host/interface address	Identifies an interface within the subnet.
Broadcast address (IPv4)	Targets all hosts in the local IPv4 broadcast domain.
Private IPv4 ranges	Used for private addressing, commonly 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.
Loopback	Used by a host to communicate with itself; IPv4 commonly 127.0.0.0/8.

Address type	Purpose
IPv6	128-bit addressing; uses hexadecimal notation and supports very large address space.

CIDR

Classless Inter-Domain Routing (CIDR) represents networks using prefix length, such as 10.10.0.0/16. CIDR enables flexible allocation and route aggregation and avoids the rigid old class A/B/C allocation model.

Subnetting example

Suppose 192.168.10.0/24 is divided into /26 subnets. Host bits = 6, so each subnet has $2^6 = 64$ addresses. The /24 can therefore be divided into 4 equal /26 blocks.

Subnet	Network	Typical broadcast
1	192.168.10.0/26	192.168.10.63
2	192.168.10.64/26	192.168.10.127
3	192.168.10.128/26	192.168.10.191
4	192.168.10.192/26	192.168.10.255

Quick Revision

IPv4 = 32 bits; IPv6 = 128 bits. Prefix length tells how many leading bits identify the network. CIDR supports flexible subnetting and route aggregation. Know /24 and /26 examples for exams.

1.8. Internetworking

Internetworking means connecting multiple independent networks so that hosts can communicate across them. The Internet is an internetwork: many autonomous networks use common protocols and routing mechanisms to exchange traffic.

Core idea

- 1 Each network may have its own local technology, such as Ethernet or Wi-Fi.
- 2 Routers connect different IP networks.
- 3 Routing determines a next hop/path toward the destination.
- 4 Packets are forwarded hop-by-hop until they reach the destination network.
- 5 The destination's local link technology finally delivers the packet to the endpoint.

Important devices

Device	Primary internetworking role
Router	Connects IP networks and forwards packets based on routing tables.
Layer-3 switch	Performs routing functions, often at high speed inside enterprise networks.
Gateway	A broader term for a boundary/forwarding point; may also perform protocol translation.
NAT device	Translates address/port information between address realms, commonly private and public IPv4.

Forwarding vs routing

Routing is the control-plane process of learning/selecting paths. **Forwarding** is the data-plane action of sending an arriving packet to the appropriate outgoing interface.

Exam Point

A routing table may contain destination prefixes, next-hop information, outgoing interfaces, and metrics. Longest-prefix matching is a fundamental IP forwarding rule.

Quick Revision

Internetworking connects multiple networks. Routers provide the key IP forwarding function. Routing chooses paths; forwarding sends individual packets according to the resulting forwarding information.

1.9. Interdomain Routing Protocols

The Internet is divided into **Autonomous Systems (ASes)**, each under a common administrative/control domain. **Interdomain routing** determines reachability between these systems.

BGP (Border Gateway Protocol) is the principal interdomain routing protocol used on the Internet. It is a path-vector protocol: routes are advertised with attributes and an AS path that helps prevent loops and supports policy decisions.

BGP concepts

Concept	Explanation
Autonomous System (AS)	A collection of IP networks/prefixes under a common routing administration and policy.
eBGP	BGP session between routers in different autonomous systems.
iBGP	BGP session between routers within the same autonomous system.
AS_PATH	Sequence of AS numbers carried with a route advertisement; used for loop prevention and policy.
NEXT_HOP	Identifies the next-hop information associated with a route.
LOCAL_PREF	Commonly used inside an AS to prefer one exit path over another.
MED	Can influence which entry point into an AS is preferred by neighboring arrangements.
Route aggregation	Combines prefixes to reduce the number of routes advertised.
Policy routing	Route choice can reflect business/administrative policy, not just shortest distance.

Simplified BGP route exchange



BGP is different from an interior gateway protocol such as OSPF or RIP because its main purpose is **interdomain** reachability and policy control.

Quick Revision

BGP = Internet-scale interdomain routing. It is path-vector based, uses AS_PATH for loop prevention, and supports policy-driven route selection. eBGP is between ASes; iBGP operates within an AS.

1.10. Common Tools for Network Analysis

Network-analysis tools help diagnose connectivity, measure performance, inspect packets, understand paths, and identify configuration problems.

Tool	Typical purpose	Example use
ping	Tests reachability and measures round-trip time using ICMP-related responses.	Check IP-related connectivity.
tracert / traceroute	Shows the sequence of hops toward a destination.	Locate where delay/loss begins.
ip / ifconfig	Displays/configures interfaces and addresses.	Inspect IP configuration.
nslookup / dig	Queries DNS.	Check name resolution and DNS records.
arp / ip neigh	Inspects local address-to-link-layer neighbor information.	Diagnose local next-hop resolution.
netstat / ss	Shows sockets and connection/listening information.	Find active services/connections.
Wireshark	Graphical packet capture and protocol analysis.	Inspect packet headers and application traffic.
tcpdump	Command-line packet capture/inspection.	Capture traffic on a selected interface.
nmap	Network discovery and service/port assessment.	Identify hosts/services in an authorized network.
iperf3	Measures network throughput and related performance.	Test link capacity.

Typical troubleshooting workflow

- 1 Check interface state and local IP configuration.
- 2 Ping the local gateway or nearby host.
- 3 Test DNS separately from raw IP connectivity.
- 4 Use traceroute to locate path-level problems.
- 5 Capture packets when behavior is unclear.
- 6 Measure throughput/latency with suitable tools and compare against expected performance.

Ethics / exam note

Tools such as packet capture and scanning should be used only on systems and networks you are authorized to analyze. In an exam, focus on the diagnostic purpose and interpretation of results.

Quick Revision

ping = reachability/RTT; traceroute = path; Wireshark/tcpdump = packet inspection; dig/nslookup = DNS; ss/netstat = sockets; ip = interface/addressing; iperf3 = performance; nmap = authorized discovery/service assessment.

Unit II — Mobile Ad Hoc Network (MANET)

Unit focus

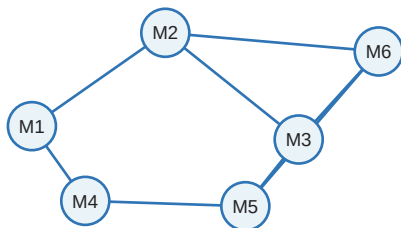
A MANET is a self-organizing wireless network of mobile nodes that communicate without relying on fixed infrastructure. This unit covers its characteristics, uses, challenges, variants, architecture, protocol stack, standards, MAC, routing, QoS, security, mobility models, and simulation tools.

2.1. MANET Overview

Mobile Ad Hoc Network (MANET) is a wireless, infrastructure-less network in which mobile nodes communicate directly when possible and can act as routers to forward packets for other nodes.

Core idea

Consider nodes A and C that are too far apart for direct communication, while B is within range of both. A can send a packet to B, and B forwards it to C. Thus, communication can be **multi-hop**.



Unlike a traditional WLAN, there may be no fixed AP coordinating the entire network. The topology can change as nodes move, links weaken, or nodes join/leave.

MANET vs infrastructure WLAN

Feature	Infrastructure WLAN	MANET
Central infrastructure	AP/base station normally present.	Not required.
Routing	Often through infrastructure gateway/AP.	Nodes may route for each other.
Topology	Relatively stable within coverage.	Potentially dynamic.
Deployment	Requires infrastructure planning.	Can be rapidly deployed.
Failure pattern	Infrastructure can be a bottleneck/critical point.	Distributed, but node/link failures are common.

Quick Revision

MANET = mobile + wireless + self-organizing + multi-hop + infrastructure-less. Every node may simultaneously behave as a host and router.

2.2. Characteristics

MANETs have properties that make them fundamentally different from conventional wired networks.

Characteristic	Meaning / implication
Dynamic topology	Node movement changes links and routes.
Infrastructure-less	No fixed AP/base station is required for basic operation.
Multi-hop communication	Packets may traverse multiple mobile nodes.
Distributed operation	Control is shared among nodes rather than relying on one central controller.
Limited bandwidth	Wireless spectrum is shared and often less predictable than wired capacity.
Variable link quality	Interference, fading, obstacles, and distance affect links.
Energy constraints	Battery-powered nodes may need energy-efficient protocols.
Scalability concerns	Routing/control overhead can grow as the network grows.
Autonomous nodes	Nodes may have different capabilities, policies, or levels of trust.
Security exposure	Wireless broadcast and dynamic membership create additional attack surfaces.

Quick Revision

The five high-value characteristics to remember: dynamic topology, infrastructure-less operation, multi-hop forwarding, shared/variable wireless resources, and distributed control.

2.3. Applications

MANETs are useful where infrastructure is unavailable, damaged, expensive, or too slow to deploy.

- **Disaster recovery:** emergency teams can form a temporary network after infrastructure damage.
- **Military/tactical communication:** mobile units may need peer-to-peer communication in changing environments.
- **Search and rescue:** rescuers, vehicles, drones, and sensors can exchange location/status information.
- **Conferences and temporary events:** devices can communicate without depending on a pre-existing network.
- **Vehicular ad hoc networking:** vehicles can exchange safety and traffic information.
- **Wireless sensor/robotic systems:** mobile or distributed devices can coordinate sensing and movement.
- **Remote/expeditionary environments:** temporary connectivity can be created where wired infrastructure is impractical.

Exam Point

The common reason behind MANET applications is rapid, decentralized deployment under changing conditions.

Quick Revision

Think: disaster, military, rescue, vehicles, temporary events, robots/sensors, remote areas.

2.4. Key Challenges

MANETs are difficult to design because mobility and wireless constraints interact with distributed routing and security.

Challenge	Why it matters	Typical response
Route breakage	Movement can invalidate a path.	Route repair, local recovery, multipath approaches.
Bandwidth variability	Shared wireless capacity changes with contention/interference.	Adaptive protocols, efficient control.
Energy limitation	Battery depletion can remove forwarding nodes.	Energy-aware routing and MAC.
Scalability	Flooding and routing tables can become expensive.	Hierarchical, clustered, or efficient routing.
Security	Nodes and links may not be physically protected.	Authentication, encryption, trust mechanisms, intrusion detection.
Hidden/exposed terminals	Radio interference causes unfairness/collisions.	RTS/CTS and MAC coordination where appropriate.
QoS variability	Delay, jitter, loss and capacity can change quickly.	Admission/control, path selection, resource awareness.
Address/configuration	Dynamic membership complicates configuration.	Distributed address/configuration methods.
Localization	Some applications need node position.	GPS/GNSS, localization protocols, or cooperative methods.

Quick Revision

MANET design is a trade-off among route stability, control overhead, energy, bandwidth, scalability, QoS, and security.

2.5. Types of Ad Hoc Network

“Ad hoc network” is a broad concept. Different types are classified by mobility, application, device role, or deployment environment.

Type	Description
MANET	Mobile nodes form a self-organizing multi-hop network.
VANET	Vehicles communicate with vehicles and/or roadside infrastructure; mobility is constrained by roads.
WMN (Wireless Mesh Network)	Mesh nodes provide multi-hop wireless connectivity; many mesh deployments have relatively stable infrastructure.
WSN (Wireless Sensor Network)	Sensor nodes collect/forward measurements, often under energy constraints; mobility varies by deployment.
FANET	Flying Ad Hoc Network, typically involving UAVs/drones with fast 3D mobility.
Hybrid ad hoc network	Combines ad hoc links with some infrastructure or external network connectivity.

Important distinction: **MANET** emphasizes mobile, self-organizing, infrastructure-less networking. **WSN** emphasizes sensing and energy-efficient data collection; **VANET** emphasizes vehicular mobility and safety/traffic applications; **FANET** emphasizes aerial nodes and rapid 3D topology changes.

Quick Revision

MANET = general mobile ad hoc; VANET = vehicles; FANET = flying/UAV; WSN = sensors; WMN = wireless mesh. These categories can overlap in real systems.

2.6. Comparison with Other Networks

MANETs are best understood by comparing them with infrastructure WLANs, cellular networks, wired networks, and wireless sensor networks.

Feature	MANET	WLAN	Cellular	Wired LAN	WSN
Fixed infrastructure	Not required	Usually yes	Yes	Yes	Often limited/optional
Node mobility	Often high	Usually low/moderate	High	Usually low	Varies

Feature	MANET	WLAN	Cellular	Wired LAN	WSN
Multi-hop among end nodes	Common	Usually not required	Mostly infrastructure-mediated	Possible via routers/switches	Common
Power constraints	May be significant	Usually less severe	Device-dependent	Low at endpoint infrastructure	Often severe
Topology dynamics	High	Moderate	Managed cell structure	Low	Varies
Primary strength	Rapid self-organizing connectivity	Convenient local access	Wide-area mobility	High stability/capacity	Distributed sensing

Exam Tip

If asked "Why MANET instead of cellular/WLAN?", answer: MANET can operate without fixed infrastructure and can be deployed quickly, but usually sacrifices the stability, centralized management, and predictable resources available in infrastructure networks.

Quick Revision

MANET trades infrastructure and centralized control for flexibility and rapid deployment.

2.7. Architecture and Communication Models

A MANET architecture is typically **distributed**. Nodes have wireless interfaces and can serve as both end systems and routers. A node may communicate directly with a neighbor or through several intermediate nodes.

Basic multi-hop communication

- 1 Source checks whether destination is directly reachable.
- 2 If not, a route discovery/selection mechanism finds a sequence of forwarding nodes.
- 3 Packets are transmitted hop-by-hop.
- 4 Intermediate nodes forward packets using their routing/forwarding information.
- 5 If mobility breaks the route, the routing protocol repairs or replaces it.

Communication models

Model	Meaning	Example
Unicast	One source to one destination.	A sends data to C.
Broadcast	One source to all nodes in the relevant broadcast scope	Route discovery request in some protocols.
Multicast	One source to a selected group.	Group communication.
Anycast-like selection	Source communicates with one suitable member of a selected group.	Request for available service and its location.
Geocast / location-oriented	Delivery based on geographic region or node location.	Alert vehicles in a road segment.

Diagram concept

In MANETs, a logical network is better viewed as a changing graph $G(V,E)$, where V is the set of nodes and E is the set of currently usable wireless links.

Quick Revision

Architecture is decentralized. Nodes can be hosts and routers. Communication can be unicast, broadcast, multicast, or location-oriented. Multi-hop forwarding is central to MANET operation.

2.8. Protocol Stack

A protocol stack organizes networking functions into layers so that each layer provides services to the layer above and uses services from the layer below.

Simplified MANET stack

Application
Transport
Network
MAC
Physical

Layer	MANET-relevant responsibilities
Application	User/application services such as messaging, sensing, voice, control.
Transport	End-to-end delivery, reliability, flow/congestion behavior (e.g., TCP/UDP).
Network	IP addressing and MANET routing/forwarding; route discovery/maintenance in ad hoc protocols.
MAC / Data Link	Channel access, framing, error handling, neighbor communication.
Physical	Radio transmission, modulation, frequency, power, propagation.

MANET routing protocols often operate at or conceptually around the network layer, but their performance depends strongly on MAC and physical-layer behavior. For example, a route that looks short may perform poorly if its wireless links have high interference or loss.

Quick Revision

Remember the stack: Application → Transport → Network → MAC/Data Link → Physical. MANET routing lives primarily at the network layer but is tightly coupled to lower-layer wireless conditions.

2.9. Wireless Standards

Wireless standards define interoperable radio, MAC, and communication behavior. Different standards target different ranges, data rates, power budgets, and use cases.

Standard / family	Primary domain	Important exam point
IEEE 802.11 (Wi-Fi)	Wireless LAN	Defines WLAN PHY/MAC families; CSMA/CA is central to classic WLAN operation.
IEEE 802.15.1 / Bluetooth	Wireless personal-area connectivity	Short-range device/peripheral networking.
IEEE 802.15.4	Low-rate wireless personal-area / IoT PHY/MAC operation	Low power and low data rate; used by higher-layer IoT technologies.
IEEE 802.16	Broadband wireless / WiMAX family	Designed for metropolitan broadband access; historically important.
Cellular 3G/4G/5G families	Wide-area mobile communication	Operator-managed mobility and cellular coverage.
802.11 ad hoc / IBSS concept	Infrastructure-less WLAN operation	Stations communicate without a conventional AP, subject to standard.

Standard vs protocol: a standard is a documented technical specification/framework; a protocol is a set of communication rules. A standard can define several protocols, procedures, and physical options.

Quick Revision

802.11 → WLAN/Wi-Fi; Bluetooth → personal-area connectivity; 802.15.4 → low-power IoT foundation; cellular → wide-area managed mobility. Know the intended domain rather than memorizing only numbers.

2.10. Medium Access Control Techniques

Medium Access Control (MAC) determines how multiple nodes share a common communication medium. In wireless networks, simultaneous transmissions can interfere, so coordination is essential.

Major MAC approaches

Technique	Principle	Advantages	Limitations
CSMA/CA	Listen before transmitting; attempt to avoid collisions using backoff and virtual CSMA/RTS/CTS.	Collision-free and no collisions.	RTS/CTS cannot be detected as easily as in wired networks.
TDMA	Time is divided into slots assigned to nodes.	Predictable access and collision-free.	Requires synchronization and scheduling.
FDMA	Different users use different frequency bands.	Parallel transmissions.	Needs spectrum allocation; can waste capacity if not used.
CDMA / spread-spectrum	Users share spectrum using coding/spreading techniques.	Multiple users can share a band.	More complex signal processing and power-control.
Polling	A controller gives nodes permission to transmit.	Controlled and orderly.	Central controller creates overhead/dependency.
RTS/CTS	Optional control handshake reserves the medium before data transmission.	Can reduce hidden-terminal collisions.	Additional control overhead and is not always beneficial.

CSMA/CA simplified flow

- 1 Sense the channel.
- 2 If the channel is busy, defer.
- 3 If idle, wait according to the contention/backoff rules.
- 4 Transmit when permitted.
- 5 Use acknowledgment where the MAC requires it.
- 6 If no acknowledgment is received, retry after another backoff attempt.

Hidden terminal problem

Suppose A and C cannot hear each other but both can reach B. A and C may transmit simultaneously toward B, causing a collision at B. RTS/CTS can help in some scenarios by informing nearby nodes of a planned transmission.

Quick Revision

MAC answers “who transmits when?” CSMA/CA is contention-based; TDMA/FDMA are scheduled/resource-partitioning approaches. RTS/CTS can mitigate hidden-terminal effects but costs extra overhead.

2.11. Routing Protocols

MANET routing is challenging because routes can break frequently. Routing protocols are commonly classified as **proactive**, **reactive**, and **hybrid**.

Category	Basic idea	Examples	Main trade-off
Proactive (table-driven)	Maintain routes to destinations continuously or periodically.	SD-WAN, DSR	Low route-discovery delay, but periodic updates.
Reactive (on-demand)	Discover a route only when a source needs it.	AODV, DSR	Lower idle overhead, but route discovery delay.

Category	Basic idea	Examples	Main trade-off
Hybrid	Combines proactive knowledge in part of the network with on-demand discovery attempts	OLSR	Attempts to balance overhead and delay

AODV — Ad hoc On-Demand Distance Vector

AODV discovers routes when needed. A source that lacks a route broadcasts a **Route Request (RREQ)**. A node that knows a valid route or the destination can generate a **Route Reply (RREP)**. Sequence numbers help maintain freshness and avoid certain routing loops. **RERR** messages can report broken routes.

AODV simplified steps

- 1 Source checks routing table.
- 2 If no valid route exists, broadcast RREQ.
- 3 Intermediate nodes record reverse-path information and forward the RREQ when appropriate.
- 4 Destination or suitable intermediate node sends RREP back.
- 5 Source installs the route and sends data.
- 6 If a link breaks, route error/repair mechanisms are used.

DSR — Dynamic Source Routing

DSR is also reactive but uses **source routing**: the packet can carry the complete route in its header. It uses route discovery and route maintenance, and can maintain route caches.

OLSR — Optimized Link State Routing

OLSR is proactive and uses **Multipoint Relays (MPRs)** to reduce flooding overhead when disseminating link-state information.

Routing comparison

Feature	AODV	DSR	OLSR
Style	Reactive	Reactive	Proactive
Route representation	Next-hop routing table	Source route in packet	Routing tables from link-state information
Control overhead	On demand + maintenance	On demand + route cache	Periodic
Good fit	Networks where demand is intermittent and topology changes are frequent	Small ad hoc networks; environments where source routing is practical	Networks where routing is continuously available

Exam Point

The key distinction is not simply the acronym: AODV = on-demand distance-vector; DSR = on-demand source routing; OLSR = proactive link-state with MPR optimization.

Quick Revision

Proactive keeps routes ready; reactive discovers routes when needed; hybrid combines both. AODV, DSR, and OLSR are the classic protocols to compare.

2.12. Quality of Service (QoS)

Quality of Service (QoS) describes the performance characteristics experienced by network traffic. In MANETs, QoS is difficult because wireless capacity and routes can change dynamically.

Metric	Meaning	Why important
Bandwidth / throughput	Rate of data delivery.	Important for high-data applications.

Metric	Meaning	Why important
Delay / latency	Time taken for delivery.	Critical for interactive/real-time traffic.
Jitter	Variation in packet delay.	Important for voice/video.
Packet loss	Fraction of packets not successfully delivered.	Affects reliability and application quality.
Reliability	Probability/consistency of successful delivery.	Important for control and mission-critical traffic.
Availability	How consistently the service is reachable.	Mobility and failures can reduce it.

QoS-aware routing may select a path using metrics such as delay, available bandwidth, hop count, energy, link stability, or a weighted combination.

Example path-selection idea

Suppose two routes exist: Route A has fewer hops but unstable links; Route B has one extra hop but better link quality and lower measured delay. A QoS-aware protocol may prefer B even though hop count alone would choose A.

A generic cost function can be written as:

$$\text{Cost} = w_{\text{Delay}} \cdot \text{Delay} + w_{\text{Loss}} \cdot \text{Loss} + w_{\text{EnergyCost}} \cdot \text{EnergyCost} + w_{\text{HopCount}} \cdot \text{HopCount}$$

where $w_{\text{Delay}}, \dots, w_{\text{HopCount}}$ are application-dependent weights. This is a conceptual model; real protocols use specific metrics and constraints.

Quick Revision

QoS is multidimensional. The four easiest metrics to remember are bandwidth, delay, jitter, and packet loss. MANET QoS is hard because mobility changes path and link conditions.

2.13. Security

MANET security protects **confidentiality, integrity, availability, authentication, and non-repudiation/accountability** as required by the application. The wireless and decentralized nature of MANETs increases the attack surface.

Threat / attack	Idea	Impact
Eavesdropping	Attacker listens to wireless transmissions.	Loss of confidentiality.
Spoofing / impersonation	Attacker pretends to be another node.	Unauthorized access, false routing information.
Blackhole	Malicious node falsely advertises a strong route and then drops traffic.	Packet loss/disruption.
Grayhole	Node selectively drops packets.	Harder to detect than constant dropping.
Wormhole	Colluding nodes create a tunnel and replay packets at another location.	Distortion of routing/topology perception.
Sybil	One node presents multiple identities.	Confuses trust/routing mechanisms.
Denial of Service (DoS)	Attempts to exhaust bandwidth, energy, processing, or routing resources.	Reduces availability.
Jamming	Interferes with the radio channel.	Prevents or degrades communication.
Routing attacks	Manipulates route advertisements or control packets.	Wrong routes, loops, traffic interception/disruption.

Security mechanisms

- **Authentication:** verify the identity or authority of communicating nodes.
- **Encryption:** protect confidentiality of data in transit.

- **Integrity protection:** detect unauthorized modification using cryptographic integrity mechanisms.
- **Secure routing:** authenticate or validate routing information and detect suspicious behavior.
- **Intrusion detection:** identify abnormal node/network behavior.
- **Trust/reputation mechanisms:** use observed behavior to evaluate forwarding reliability.
- **Key management:** distribute, rotate, revoke, and protect cryptographic keys.

Exam Point

Security is not only encryption. A MANET must also protect routing/control information and cope with compromised or selfish nodes.

Quick Revision

CIA + authentication are the core security goals. Remember blackhole, grayhole, wormhole, Sybil, eavesdropping, spoofing, DoS/jamming, and routing attacks.

2.14. Mobility Models and Simulation Tools

A **mobility model** mathematically or algorithmically describes how nodes move over time. Mobility strongly affects MANET topology, route lifetime, link availability, and protocol performance.

Mobility model	Basic behavior	Use / characteristic
Random Waypoint	Node chooses a random destination and speed, moves there, pauses, then repeats.	Simple baseline; common in MANET studies.
Random Walk	Movement changes in random directions/steps.	Simple stochastic mobility.
Gauss-Markov	Uses correlation in velocity/direction to produce smoother movement.	More realistic than fully independent random movement in some cases.
Manhattan Grid	Movement follows grid-like roads with constrained turns.	Useful for urban/vehicular-like scenarios.
Reference Point Group Mobility	Nodes move around a reference point/group center.	Useful for teams or coordinated groups.
Pursue mobility	Nodes move toward a target/reference node.	Useful for chase/target scenarios.
Gauss-Markov / correlated models	Movement has temporal correlation.	Can model smoother trajectories.

Why mobility models matter

A routing protocol tested under very high random mobility may behave differently from the same protocol in a road-constrained or group-mobility scenario. Therefore, simulation results are meaningful only when the mobility and traffic assumptions are clearly stated.

Common simulation tools

Tool	Typical use
NS-2	Classic discrete-event network simulator; widely used in academic MANET literature.
NS-3	Modern discrete-event network simulator used for IP, wireless, and network-protocol studies.
OMNeT++	Modular discrete-event simulation framework; often used with networking models/frameworks.
MATLAB	Useful for algorithm development, mathematical modeling, and result analysis; network simulation often relies on external tools.
GloMoSim	Historically used for scalable wireless/mobile network simulation.
SUMO	Traffic simulator commonly coupled with network simulators for vehicular mobility studies.

Typical MANET simulation workflow

- 1 Define network size, radio model, propagation assumptions, and node count.
- 2 Select a mobility model and mobility parameters.
- 3 Select traffic/application model.
- 4 Configure MAC, routing protocol, transport, and energy assumptions.
- 5 Run multiple scenarios/seeds where appropriate.
- 6 Collect metrics such as packet delivery ratio, throughput, delay, routing overhead, and energy consumption.
- 7 Compare results and explain why mobility, load, or protocol behavior caused the observed differences.

Common evaluation metrics

Metric	Formula / interpretation
Packet Delivery Ratio (PDR)	$\text{PDR} = \text{packets received} / \text{packets sent} \times 100\%$.
Throughput	Useful data delivered per unit time, often bit/s.
Average end-to-end delay	Mean time from packet generation to successful reception.
Routing overhead	Control traffic or routing-control transmissions relative to delivered data, depending on the study's definition.
Energy consumption	Energy used by nodes during communication/computation; important for battery-powered systems.

Quick Revision

Mobility models define movement; simulators execute network behavior. Random Waypoint is a classic baseline, while group/grid/vehicular models can better represent specific scenarios. Always state assumptions and evaluation metrics.

Final Revision Section

A. Important Definitions

Definition	Exam-ready meaning
Computer network	Interconnected nodes that exchange data and share resources using communication protocols.
Wireless network	Network in which at least part of communication uses a wireless transmission medium.
Protocol	Rules governing communication between networked entities.
Topology	Arrangement of network nodes and communication links.
Internetworking	Connecting multiple networks so they can communicate using common/interworking protocols.
Autonomous System	A collection of networks/prefixes under a common administrative routing policy.
BGP	The principal Internet interdomain routing protocol; a path-vector protocol supporting policy-based route selection.
MANET	A mobile, self-organizing, infrastructure-less wireless network in which nodes may forward traffic for one another.
MAC	Mechanisms controlling access by multiple nodes to a shared communication medium.
QoS	Performance characteristics such as delay, jitter, throughput, loss, and reliability experienced by network traffic.
Mobility model	A model describing how nodes move over time in a simulation or analytical study.

B. Important Formulas

Concept	Formula / rule
IPv4 subnet addresses	$2^{(32 - \text{prefix length})}$
Traditional usable IPv4 hosts per subnet	$2^h - 2$, where h = host bits
Packet Delivery Ratio	$\text{PDR} = \text{received packets} / \text{sent packets} \times 100\%$
Throughput	Useful data delivered / time
Generic weighted routing cost	$\text{Cost} = \sum(w_i \times \text{metric}_i)$
Link/path intuition	More hops can increase delay and failure opportunities, but a shorter-hop route is not automatically better.

C. Important Comparisons

Comparison	Key difference
Circuit vs packet switching	Circuit reserves dedicated resources; packet switching shares resources dynamically.

Comparison	Key difference
Routing vs forwarding	Routing selects/learns paths; forwarding sends packets using forwarding information.
AP vs router	AP provides wireless LAN access; router connects IP networks. A consumer wireless router often functions as both.
Proactive vs reactive MANET routing	Proactive maintains routes continuously; reactive discovers routes on demand.
AODV vs DSR	Both reactive; AODV uses distance-vector-style next-hop tables, while DSR uses source routing.
OLSR vs AODV	OLSR is proactive link-state with MPR optimization; AODV is reactive/on-demand.
MANET vs WLAN	MANET can operate without fixed infrastructure and uses multi-hop peer forwarding; WLAN normally requires infrastructure.
MANET vs cellular	MANET is decentralized/infrastructure-less; cellular relies on operator infrastructure and managed network.
CSMA/CA vs TDMA	CSMA/CA contends for the medium; TDMA assigns time slots.
Blackhole vs grayhole	Blackhole can drop essentially all selected traffic; grayhole drops selectively.

D. Important Diagrams / Concepts to Draw in Exams

- **Star topology:** all end nodes connect to a central device.
- **Mesh topology:** multiple redundant links between nodes.
- **MANET multi-hop graph:** show source → intermediate nodes → destination.
- **Protocol stack:** Application → Transport → Network → MAC/Data Link → Physical.
- **Hidden terminal:** A and C cannot hear each other but both transmit to B.
- **AODV flow:** RREQ → intermediate forwarding → RREP → data → route error/repair if needed.
- **BGP:** show two or more autonomous systems exchanging reachability information.
- **Simulation workflow:** topology + mobility + traffic + protocol → simulation → metrics → comparison.

E. Frequently Testable Concepts

- 1 Define MANET and explain its characteristics.
- 2 Explain MANET applications and major challenges.
- 3 Differentiate proactive, reactive, and hybrid routing protocols.
- 4 Explain AODV with RREQ, RREP, and RERR.
- 5 Compare AODV, DSR, and OLSR.
- 6 Explain CSMA/CA and the hidden-terminal problem.
- 7 Explain QoS metrics in MANETs.
- 8 Discuss MANET security threats and countermeasures.
- 9 Explain mobility models and why they affect simulation results.
- 10 Explain IPv4 addressing, subnetting, and CIDR.
- 11 Differentiate routing and forwarding.
- 12 Explain BGP and interdomain routing.
- 13 Describe common network-analysis tools and their purposes.
- 14 Compare MANET with WLAN, cellular, and WSN.

F. Rapid Revision — Unit I

Unit I in one page

Networking: nodes + links + protocols. **Wireless:** uses electromagnetic transmission. **Topologies:** bus, star, ring, mesh, tree, hybrid. **Switching:** circuit, packet, message. **Protocols:** Ethernet/Wi-Fi, IP, TCP/UDP, DNS, DHCP, BGP. **IP:** IPv4 = 32-bit, IPv6 = 128-bit; CIDR uses prefix length. **Internetworking:** routers connect IP networks. **BGP:** interdomain path-vector routing. **Tools:** ping, traceroute, ip, dig/nslookup, Wireshark, tcpdump, ss/netstat, iperf3, nmap.

G. Rapid Revision — Unit II

Unit II in one page

MANET: mobile, wireless, infrastructure-less, self-organizing, multi-hop. **Characteristics:** dynamic topology, shared bandwidth, energy constraints, distributed control. **Applications:** disaster, military, rescue, vehicular, temporary/remote networks. **Types:** MANET, VANET, FANET, WSN, WMN, hybrid. **Stack:** Application → Transport → Network → MAC → Physical. **MAC:** CSMA/CA, TDMA, FDMA, RTS/CTS. **Routing:** proactive (OLSR/DSDV), reactive (AODV/DSR), hybrid (ZRP). **QoS:** throughput, delay, jitter, loss, reliability. **Security:** eavesdropping, spoofing, blackhole, grayhole, wormhole, Sybil, DoS/jamming. **Mobility:** Random Waypoint, Random Walk, Gauss-Markov, Manhattan, group mobility. **Simulation:** NS-2, NS-3, OMNeT++, MATLAB, GloMoSim, SUMO.

Last-minute exam strategy

For a 5–10 mark answer: start with a definition, draw a simple labeled diagram if applicable, explain working in numbered steps, list advantages/limitations or a comparison, then finish with 3–5 key points/applications. For routing and MAC questions, explicitly name the control messages or access steps.